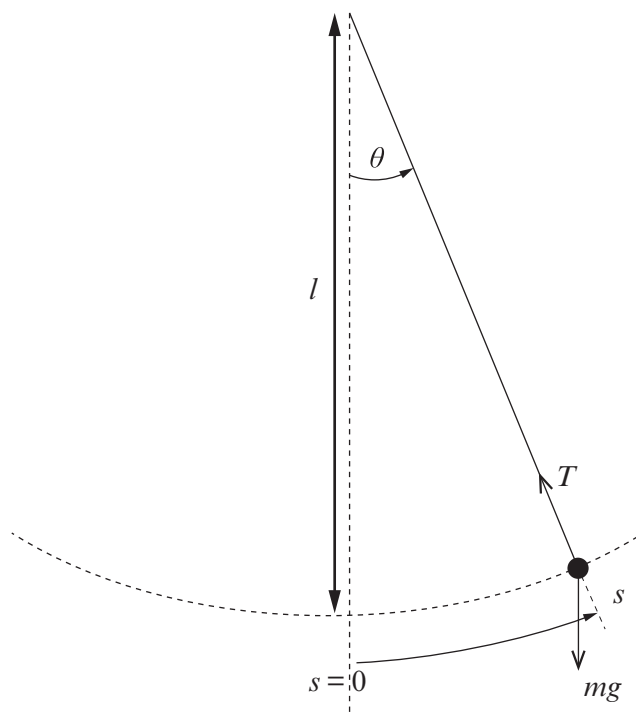


6. (a) The pendulum in the figure below has a small bob of mass, m , suspended at the bottom of a light string of length, l . The string is shown at an angle, θ , to the vertical and the mass, m , is at a distance, s , along the arc. The forces acting on the mass are shown.



- (i) Name the **two** forces acting on the mass.

[1]

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.....

- (ii) By considering these forces show that:

resultant force component on the mass along the arc = $-mg \sin \theta$

You may add to the diagram if you wish.

[3]

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- (iii) If the oscillation is small so that $\sin \theta \approx \theta$ show in clear steps that the acceleration along the arc may be written as: [2]

$$\text{acceleration} = -\frac{gs}{l}$$

.....

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- (iv) Discuss whether the equation in part (a)(iii) satisfies the definition of simple harmonic motion. [2]

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- (b) A small mass oscillates at the lower end of a pendulum of length 1.20 m.

- (i) Determine its:

I. period;

[2]

.....

.....

.....

II. frequency.

[1]

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- (ii) If the maximum displacement angle of the mass is 0.067 rad, justify the use of simple harmonic motion in part (b)(i). [2]

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